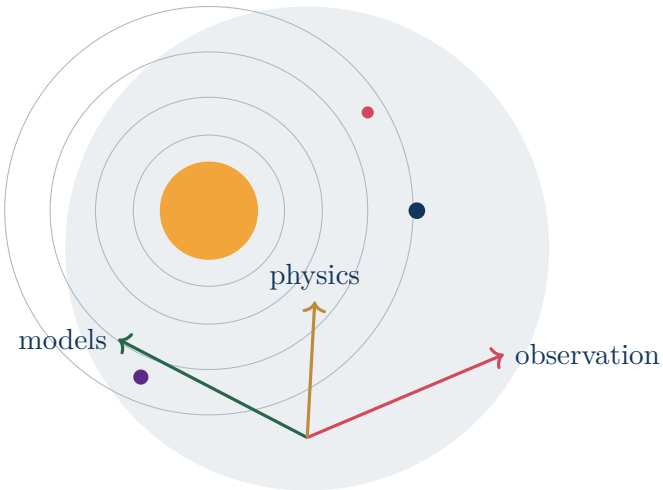
Astrophysics Lecture Notes

From Light and Gravity to Stars, Galaxies, and Cosmology



Lecture notes prepared for a mathematical and physical introduction to astrophysics. May 16, 2026

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1 What Astrophysics Is

Astrophysics is the use of physics to understand astronomical objects. Astronomy asks: *what do we observe in the sky?* Astrophysics asks: *what physical laws explain these observations?*

The central difficulty is that almost everything in astrophysics is inaccessible to laboratory manipulation. We cannot take a star into a laboratory, change its mass, and repeat the experiment. Instead, astrophysics is built from three pillars:

1. **Observation**: light, particles, gravitational waves, and sometimes direct samples such as meteorites.
2. **Physical theory**: mechanics, electromagnetism, thermodynamics, statistical physics, relativity, nuclear physics, and quantum physics.
3. **Models**: mathematical approximations that connect theory with measurable quantities.

The basic logic of astrophysics

We observe radiation or motion. From that we infer physical quantities such as temperature, mass, radius, velocity, density, luminosity, composition, age, and distance. Then we test whether a physical model can explain them simultaneously.

1.1 Objects and scales

Astrophysics covers an enormous range of length, time, mass, and energy scales. The same laws are used from planets to galaxies, but the dominant physical effects change.

Object	Typical scale	Dominant physics
Planetary system	$1 - 100\,\mathrm{AU}$	Newtonian gravity, radiation, atmospheres
Star	$10^5 - 10^7\,\mathrm{km}$	gravity, pressure, nuclear reactions, radiation
White dwarf	Earth-sized	degenerate matter, gravity
Neutron star	$\sim 10\,\mathrm{km}$	nuclear density matter, relativity
Galaxy	$10^3 - 10^5\,\mathrm{pc}$	gravity, dark matter, stellar populations
Universe	gigaparsecs	general relativity, expansion, cosmology

1.2 A few constants and units

Quantity	Symbol	Approximate value
Speed of light	c	$3.00 \times 10^8\,\mathrm{m\,s^{-1}}$
Gravitational constant	G	$6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$
Boltzmann constant	k_B	$1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
Planck constant	h	$6.63 \times 10^{-34}\,\mathrm{J\,s}$
Solar mass	$M_\odot$	$1.99 \times 10^{30}\,\mathrm{kg}$
Solar radius	$R_\odot$	$6.96 \times 10^8\,\mathrm{m}$
Solar luminosity	$L_\odot$	$3.83 \times 10^{26}\,\mathrm{W}$
Astronomical unit	AU	$1.50 \times 10^{11}\,\mathrm{m}$
Parsec	pc	$3.09 \times 10^{16}\,\mathrm{m}$

Parsec

One parsec is the distance from which one astronomical unit subtends an angle of one arcsecond. Numerically,

$$1\,\mathrm{pc} \approx 3.26\,\text{light years}.$$

The parsec is natural in astronomy because it is tied to parallax measurements.

2 Light as the Main Source of Information

Most astrophysical information arrives as electromagnetic radiation. Therefore, before discussing stars or galaxies, we must understand how light carries information.

2.1 Flux, luminosity, and distance

The **luminosity** L of an object is the total power emitted by it. The **flux** F is the power received per unit area by an observer.

If an object emits isotropically, its radiation spreads over the surface of a sphere of radius d. The surface area is $4\pi d^2$. Hence

$$F = \frac{L}{4\pi d^2}.$$

Inverse-square law

Flux decreases as distance squared. If a star is twice as far away, its observed flux is four times smaller.

Luminosity from flux and distance

Suppose a star is at distance $d = 10\,\mathrm{pc}$ and has measured flux

$$F = 3.2 \times 10^{-10}\,\mathrm{W\,m^{-2}}.$$

Since $10\,\mathrm{pc} \approx 3.09 \times 10^{17}\,\mathrm{m}$,

$$L = 4\pi d^2 F = 4\pi(3.09 \times 10^{17})^2(3.2 \times 10^{-10}) \approx 3.8 \times 10^{26}\,\mathrm{W}.$$

This is approximately the luminosity of the Sun.

2.2 Magnitudes

Astronomers often use magnitudes instead of fluxes. The magnitude scale is logarithmic. If two objects have fluxes F_1 and F_2, their apparent magnitudes satisfy

$$m_1 - m_2 = -2.5 \log_{10}\left(\frac{F_1}{F_2}\right).$$

A smaller magnitude means a brighter object.

Absolute magnitude is the apparent magnitude an object would have if it were placed at $10\,\mathrm{pc}$. The distance modulus is

$$m - M = 5 \log_{10}\left(\frac{d}{10\,\mathrm{pc}}\right).$$

Distance modulus

If a star has apparent magnitude $m = 12$ and absolute magnitude $M = 2$, then

$$12 - 2 = 5 \log_{10} \left(\frac{d}{10\,\mathrm{pc}} \right).$$

Therefore

$$2 = \log_{10} \left(\frac{d}{10\,\mathrm{pc}} \right),$$

so

$$\frac{d}{10\,\mathrm{pc}} = 100, \qquad d = 1000\,\mathrm{pc}.$$

2.3 Blackbody radiation

A blackbody is an idealized object that absorbs all incident radiation and emits a spectrum determined only by its temperature. The spectral radiance as a function of frequency is given by Planck's law:

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1}.$$

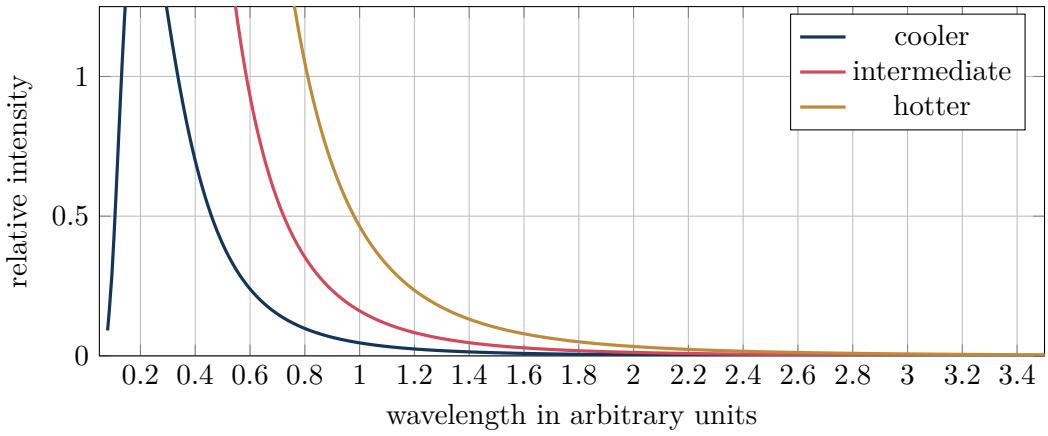
Important consequences:

- hotter objects emit more radiation at all wavelengths;
- hotter objects peak at shorter wavelengths;
- the total emitted power per unit area is σT^4.

The Stefan-Boltzmann law for a spherical star is

$$L = 4\pi R^2 \sigma T_{\mathrm{eff}}^4.$$

Blackbody-like curves: hotter means brighter and bluer



2.4 Spectral lines

A spectrum is not only a continuum. It often contains absorption or emission lines. Spectral lines are fingerprints of atoms, ions, and molecules.

A line can tell us:

- composition: which elements are present;
- temperature and ionization state;
- density and pressure through line shapes;
- velocity through Doppler shifts;
- magnetic fields through Zeeman splitting.

For non-relativistic velocities, the Doppler relation is approximately

$$\frac{\Delta\lambda}{\lambda_0} \approx \frac{v_r}{c},$$

where v_r is radial velocity. A positive shift to longer wavelength means recession.

Radial velocity from a spectral line

A line with rest wavelength $\lambda_0 = 500\,\mathrm{nm}$ is observed at $\lambda = 501\,\mathrm{nm}$. Then

$$\frac{\Delta\lambda}{\lambda_0} = \frac{1}{500} = 0.002.$$

Thus

$$v_r \approx 0.002c \approx 600\,\mathrm{km\,s^{-1}}.$$

The object is moving away from us along the line of sight.

3 Gravity, Orbits, and Mass Measurement

Gravity is the organizing force of astrophysics. It binds planets to stars, stars to galaxies, and galaxies to clusters.

3.1 Newtonian gravity

Two point masses attract each other with force

$$F = G\frac{m_1 m_2}{r^2}.$$

For a small body orbiting a massive object M, the acceleration is

$$a = \frac{GM}{r^2}.$$

3.2 Circular orbits

For a circular orbit of radius r, centripetal acceleration is

$$a_c = \frac{v^2}{r}.$$

Equating this to gravitational acceleration gives

$$\frac{v^2}{r} = \frac{GM}{r^2},$$

so

$$v = \sqrt{\frac{GM}{r}}.$$

The orbital period is

$$P = \frac{2\pi r}{v} = 2\pi\sqrt{\frac{r^3}{GM}}.$$

Therefore

$$P^2 = \frac{4\pi^2}{GM} r^3.$$

This is Kepler's third law in Newtonian form.

Mass from an orbit

A star orbits the center of a compact system at radius $r = 1.5 \times 10^{14}\,\mathrm{m}$ with speed $v = 3.0 \times 10^6\,\mathrm{m\,s^{-1}}$. Estimate the central mass.

$$M = \frac{v^2 r}{G} = \frac{(3.0 \times 10^6)^2 (1.5 \times 10^{14})}{6.67 \times 10^{-11}} \approx 2.0 \times 10^{37}\,\mathrm{kg}.$$

In solar masses:

$$M \approx \frac{2.0 \times 10^{37}}{1.99 \times 10^{30}} \approx 10^7\,M_\odot.$$

Orbital dynamics is one of the strongest ways to infer masses in astrophysics.

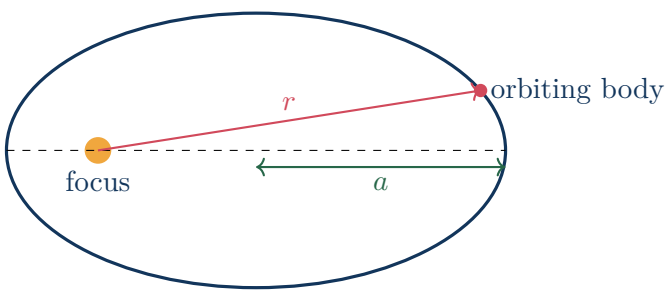
3.3 Elliptical orbits

Bound Keplerian orbits are ellipses with the central mass at one focus. The semi-major axis a determines the period:

$$P^2 = \frac{4\pi^2}{G(M+m)} a^3.$$

If $m \ll M$, this becomes

$$P^2 \approx \frac{4\pi^2}{GM} a^3.$$



A bound two-body orbit is an ellipse.
The period is controlled by the semi-major axis.

3.4 Virial theorem

For many gravitationally bound systems in equilibrium, such as star clusters, the virial theorem states

$$2K + U = 0,$$

where K is total kinetic energy and $U < 0$ is gravitational potential energy.

For order-of-magnitude estimates,

$$K \sim \frac{1}{2}Mv^2, \qquad U \sim -\frac{GM^2}{R}.$$

Thus

$$Mv^2 \sim \frac{GM^2}{R},$$

so

$$M \sim \frac{Rv^2}{G}.$$

Why the virial theorem matters

It allows us to estimate the mass of a gravitationally bound object from its size and internal velocities. This is used for star clusters, galaxies, and galaxy clusters.

4 Radiation, Temperature, and Matter

Astrophysical gas is often hot, ionized, and dilute. Its behavior is described by thermodynamics, statistical physics, and radiation physics.

4.1 Ideal gas pressure

For a gas with mass density ρ, temperature T, and mean molecular weight μ, pressure is

$$P = \frac{\rho k_B T}{\mu m_p},$$

where m_p is the proton mass. The parameter μ measures average particle mass in units of m_p.

For a fully ionized hydrogen gas, one proton and one electron are produced per hydrogen atom. The number of particles doubles, so $\mu \approx 1/2$.

4.2 Hydrostatic equilibrium

A star does not immediately collapse because pressure gradients oppose gravity. Consider a thin spherical shell inside a star. The inward gravitational force is balanced by the outward pressure difference.

The hydrostatic equilibrium equation is

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{Gm(r)\rho(r)}{r^2},$$

where $m(r)$ is the mass inside radius r.

The mass conservation equation is

$$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2 \rho(r).$$

Hydrostatic equilibrium

A star in hydrostatic equilibrium is not static because nothing happens inside it. It is static because gravity and pressure forces balance at every radius.

4.3 Radiative diffusion

Photons in stellar interiors do not fly straight outward. They scatter many times. Energy transport by radiation behaves like diffusion.

A simplified radiative temperature gradient is

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3\kappa\rho L(r)}{16\pi a c T^3 r^2},$$

where κ is opacity and a is the radiation constant.

High opacity means radiation has difficulty escaping, producing a steeper temperature gradient.

4.4 Radiation pressure

Radiation carries momentum. The pressure of blackbody radiation is

$$P_{\mathrm{rad}} = \frac{1}{3} a T^4.$$

In very massive stars, radiation pressure becomes important. This affects stability and mass loss.

5 Stars: Basic Observable Properties

A star can often be characterized approximately by mass M, radius R, luminosity L, effective temperature T_{eff}, and composition.

5.1 Effective temperature

The effective temperature is defined by pretending the star radiates as a blackbody with the same total luminosity:

$$L = 4\pi R^2 \sigma T_{\mathrm{eff}}^4.$$

Thus

$$T_{\mathrm{eff}} = \left(\frac{L}{4\pi R^2 \sigma}\right)^{1/4}.$$

Radius from luminosity and temperature

A star has $L = 100 L_\odot$ and $T_{\mathrm{eff}} = 2T_\odot$. Since

$$\frac{L}{L_\odot} = \left(\frac{R}{R_\odot}\right)^2 \left(\frac{T}{T_\odot}\right)^4,$$

we get

$$100 = \left(\frac{R}{R_\odot}\right)^2 (2)^4 = 16 \left(\frac{R}{R_\odot}\right)^2.$$

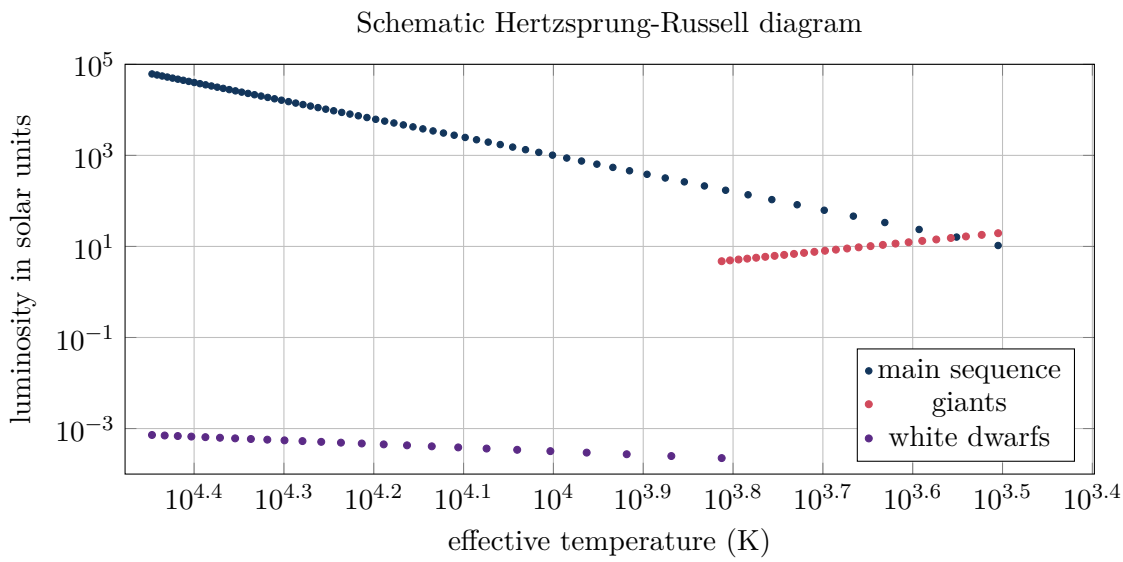
Therefore

$$\frac{R}{R_\odot} = \sqrt{\frac{100}{16}} = 2.5.$$

The star is hotter and larger than the Sun.

5.2 The Hertzsprung-Russell diagram

The H-R diagram plots luminosity against effective temperature or color. Temperature is conventionally shown decreasing to the right.



Main regions:

- **Main sequence**: stars burning hydrogen in their cores.
- **Red giants**: cool surface, large radius, high luminosity.
- **White dwarfs**: hot but faint because they are small.

5.3 Mass-luminosity relation

For main-sequence stars, luminosity increases strongly with mass. A rough approximation is

$$L \propto M^{\alpha},$$

where $\alpha \approx 3 - 4$ for stars roughly similar to the Sun.

This implies that massive stars burn fuel much faster. Their lifetimes are shorter even though they have more fuel.

If available fuel scales as M and luminosity scales as M^{α}, then lifetime scales as

$$tau \sim \frac{M}{L} \sim M^{1-\alpha}.$$

For $\alpha = 3.5$,

$$tau \sim M^{-2.5}.$$

Why massive stars live short lives

Compare a $10M_\odot$ star with the Sun. If $L \propto M^{3.5}$, then

$$\frac{L}{L_\odot} \approx 10^{3.5} \approx 3160.$$

Its lifetime relative to the Sun is approximately

$$\frac{\tau}{\tau_\odot} \approx \frac{10}{3160} \approx 0.0032.$$

If the Sun lives about 10^{10} years on the main sequence, this star lives only about 3×10^7 years.

6 Stellar Structure: The Four Basic Equations

A simple stellar model is governed by four coupled differential equations.

6.1 Mass conservation

$$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2 \rho.$$

This states that the mass added when radius increases by $\mathrm{d}r$ is density times shell volume.

6.2 Hydrostatic equilibrium

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{Gm\rho}{r^2}.$$

Pressure decreases outward because it must support the weight of overlying material.

6.3 Energy generation

$$\frac{\mathrm{d}L}{\mathrm{d}r} = 4\pi r^2 \rho \epsilon,$$

where ϵ is energy generation rate per unit mass.

6.4 Energy transport

Energy may be transported by radiation, convection, or conduction. In radiative regions,

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3\kappa\rho L}{16\pi a c T^3 r^2}.$$

Together with an equation of state, opacity law, and nuclear reaction rates, these equations define a stellar model.

Why stellar structure is difficult

The equations are simple to write but difficult to solve because pressure, density, temperature, opacity, luminosity, and nuclear reactions all depend on each other.

6.5 Order-of-magnitude central temperature

For a star of mass M and radius R, hydrostatic equilibrium gives approximately

$$\frac{P_c}{R} \sim \frac{GM\rho}{R^2}.$$

Since $\rho \sim M/R^3$,

$$P_c \sim \frac{GM^2}{R^4}.$$

For ideal gas pressure,

$$P_c \sim \frac{\rho k_B T_c}{\mu m_p} \sim \frac{M}{R^3}\frac{k_B T_c}{\mu m_p}.$$

Equating both estimates gives

$$T_c \sim \frac{GM\mu m_p}{k_B R}.$$

This explains why massive compact objects become hot enough for nuclear reactions.

7 Nuclear Energy in Stars

Stars shine because nuclear reactions convert mass into energy. The basic relation is

$$E = mc^2.$$

Hydrogen fusion converts four protons into a helium nucleus plus energy.

7.1 The proton-proton chain

The proton-proton chain dominates in stars like the Sun. The net result is approximately

$$4p \to {}^4\mathrm{He} + 2e^+ + 2\nu_e + \gamma + \text{energy}.$$

Only about 0.7% of the mass is converted to energy, but because c^2 is enormous, this is enough to power stars for billions of years.

7.2 The CNO cycle

In more massive stars, hydrogen burning is dominated by the CNO cycle, in which carbon, nitrogen, and oxygen act as catalysts. The net result is still hydrogen to helium, but the temperature dependence is much stronger.

7.3 Why fusion requires high temperature

Protons repel each other electrically. To fuse, they must come close enough for the strong nuclear force to act. High temperature gives particles higher kinetic energy, and quantum tunneling allows fusion even when classical energy is insufficient.

Common misconception

Stars do not burn chemically. Chemical burning cannot produce enough energy to power the Sun for billions of years. Stellar energy is nuclear.

8 Stellar Evolution

A star's evolution is largely determined by its initial mass.

8.1 Formation

Stars form in cold molecular clouds. Gravity pulls gas inward. As collapse proceeds, gravitational potential energy is converted into heat. If the core reaches sufficient temperature and density, hydrogen fusion begins.

A collapsing object becomes a main-sequence star when nuclear energy generation balances radiative losses and hydrostatic equilibrium is established.

8.2 Main sequence

On the main sequence, a star burns hydrogen in its core. This is the longest stable phase of stellar life.

The star is stable because of a thermostat mechanism:

- if the core contracts, it heats up;
- higher temperature increases fusion rate;
- increased pressure pushes back against gravity;
- the core expands and cools.

8.3 Red giant phase

When core hydrogen is exhausted, fusion continues in a shell around the core. The outer layers expand and cool, producing a red giant.

The luminosity rises because the radius becomes large:

$$L = 4\pi R^2 \sigma T_{\text{eff}}^4.$$

Even if the surface temperature is lower, the much larger radius can make luminosity enormous.

8.4 Low-mass and high-mass endpoints

Initial mass	Main endpoint	Remnant
Low/intermediate	red giant, planetary nebula	white dwarf
Massive	supergiant, core collapse supernova	neutron star or black hole
Very massive	strong mass loss, collapse	black hole

9 White Dwarfs, Neutron Stars, and Black Holes

Compact objects are laboratories for extreme physics.

9.1 White dwarfs

A white dwarf is supported by electron degeneracy pressure, a quantum mechanical pressure that does not require heat. It is roughly Earth-sized but contains a mass comparable to the Sun.

There is a maximum mass for a non-rotating white dwarf, the Chandrasekhar mass:

$$M_{\text{Ch}} \approx 1.4 M_\odot.$$

If a white dwarf approaches this limit, it cannot remain stable as a normal white dwarf.

9.2 Neutron stars

A neutron star has mass of order $1 - 2M_\odot$ compressed into a radius of about 10 km. Density is comparable to nuclear density.

The escape speed is a large fraction of the speed of light:

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}}.$$

Escape speed from a neutron star

Let $M = 1.4M_\odot$ and $R = 12\,\text{km}$. Then

$$v_{\text{esc}} = \sqrt{\frac{2G(1.4M_\odot)}{R}}.$$

Using $M_\odot = 1.99 \times 10^{30}\,\text{kg}$,

$$v_{\text{esc}} \approx \sqrt{\frac{2(6.67 \times 10^{-11})(2.79 \times 10^{30})}{1.2 \times 10^4}} \approx 1.8 \times 10^8\,\text{m}\,\text{s}^{-1}.$$

This is about $0.6c$, so relativity is essential.

9.3 Black holes

A black hole is a region from which no light can escape. For a non-rotating black hole, the Schwarzschild radius is

$$r_s = \frac{2GM}{c^2}.$$

For the Sun's mass,

$$r_s \approx 3\,\text{km}.$$

This does not mean the Sun is a black hole. It means the Sun would become one only if compressed within about 3 km.

Event horizon

The event horizon is not a material surface. It is a causal boundary: once crossed inward, future-directed paths cannot return to the outside universe.

10 Interstellar Medium and Star Formation

The space between stars is not empty. It contains gas, dust, cosmic rays, and magnetic fields.

10.1 Phases of the interstellar medium

The interstellar medium has multiple phases:

- cold molecular clouds;
- neutral atomic gas;
- warm ionized gas;
- hot ionized gas;
- dense star-forming cores.

10.2 Jeans instability

A gas cloud collapses if gravity overcomes pressure support. A simplified form of the Jeans length is

$$\lambda_J \sim c_s \sqrt{\frac{\pi}{G\rho}},$$

where c_s is sound speed.

The corresponding Jeans mass is roughly

$$M_J \sim \rho \lambda_J^3.$$

If a cloud mass exceeds the Jeans mass, it can collapse gravitationally.

Physical meaning

High temperature raises pressure and resists collapse. High density strengthens gravity and promotes collapse.

10.3 Dust

Dust grains absorb and scatter visible light, causing extinction and reddening. They also emit infrared radiation and help shield molecular gas from ultraviolet radiation.

The observed color of an object can therefore be affected by both its intrinsic temperature and dust along the line of sight.

11 Galaxies

A galaxy is a gravitationally bound system of stars, gas, dust, dark matter, and compact remnants.

11.1 Types of galaxies

Basic morphological classes:

- **spiral galaxies**: disks, spiral arms, gas, dust, ongoing star formation;
- **elliptical galaxies**: older stellar populations, little cold gas, smooth shapes;
- **irregular galaxies**: disturbed or asymmetric systems.

11.2 Rotation curves and dark matter

For circular motion in a gravitational field,

$$v(r) = \sqrt{\frac{GM(r)}{r}}.$$

If most mass were concentrated in the bright central region, then far outside it one would expect

$$v(r) \propto r^{-1/2}.$$

But many galaxy rotation curves remain approximately flat:

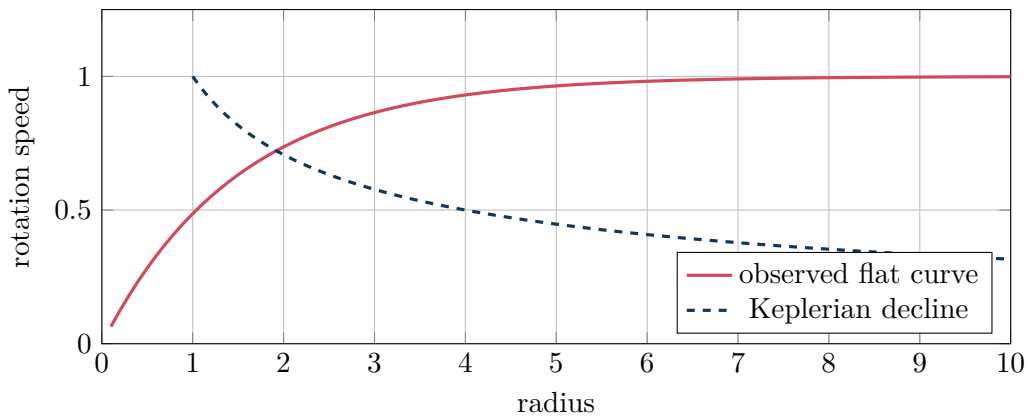
$$v(r) \approx \text{constant}.$$

This implies

$$M(r) \propto r.$$

The mass continues increasing with radius even where little light is observed. This is one of the main lines of evidence for dark matter.

Schematic galaxy rotation curve



11.3 The virial mass of a galaxy

If a galaxy has characteristic radius R and velocity dispersion σ_v, the virial theorem gives

$$M \sim \frac{R\sigma_v^2}{G}.$$

When this mass exceeds the mass inferred from visible stars and gas, the difference is attributed to dark matter.

12 Cosmology: The Universe as a Physical System

Cosmology studies the universe on the largest scales.

12.1 Expansion of the universe

Hubble's law states approximately

$$v = H_0 d,$$

where v is recession speed, d is distance, and H_0 is the Hubble constant.

For small redshifts,

$$z \approx \frac{v}{c}.$$

Thus

$$cz \approx H_0 d.$$

Distance from redshift at small redshift

Assume $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. If a galaxy has $z = 0.01$, then

$$v \approx cz = (3.0 \times 10^5\,\mathrm{km\,s^{-1}})(0.01) = 3000\,\mathrm{km\,s^{-1}}.$$

Therefore

$$d \approx \frac{v}{H_0} = \frac{3000}{70}\,\mathrm{Mpc} \approx 43\,\mathrm{Mpc}.$$

12.2 Scale factor and redshift

The expansion of the universe is described by a scale factor $a(t)$. Redshift is related to the scale factor by

$$1 + z = \frac{a(t_0)}{a(t_{\mathrm{emit}})}.$$

If we set $a(t_0) = 1$, then

$$a(t_{\mathrm{emit}}) = \frac{1}{1+z}.$$

12.3 Friedmann equation

In general relativity, the expansion of a homogeneous and isotropic universe is governed by the Friedmann equation:

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}.$$

The terms represent matter-energy density, spatial curvature, and the cosmological constant.

In modern notation, this is often written as

$$\frac{H^2(a)}{H_0^2} = \Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_k a^{-2} + \Omega_\Lambda.$$

12.4 Cosmic microwave background

The cosmic microwave background is relic radiation from the early universe. It is nearly a perfect blackbody with temperature about 2.7 K. It provides evidence that the universe was once hot, dense, and opaque.

Small temperature anisotropies in the cosmic microwave background encode information about the composition, geometry, and early fluctuations of the universe.

13 Distances: The Cosmic Distance Ladder

Distance measurement is one of the hardest parts of astronomy.

13.1 Parallax

For nearby stars, distance can be measured by parallax. If parallax angle is p arcseconds, then

$$d(\mathrm{pc}) = \frac{1}{p(\mathrm{arcsec})}.$$

Parallax distance

A star has parallax $p = 0.05''$. Then

$$d = \frac{1}{0.05} = 20\,\mathrm{pc}.$$

13.2 Standard candles

If an object has known luminosity L, measuring flux gives distance:

$$d = \sqrt{\frac{L}{4\pi F}}.$$

Objects used in this way are called standard candles.

Examples include Cepheid variables and Type Ia supernovae, although each requires calibration and systematic corrections.

14 Observational Windows and Instruments

Different physical processes emit at different wavelengths.

Band	Typical sources	Physical meaning
Radio	cold gas, synchrotron emission	magnetic fields, neutral hydrogen
Infrared	dust, cool stars, star formation	obscured regions
Visible	stars	stellar populations
Ultraviolet	hot young stars	recent star formation
X-ray	hot gas, accretion disks	high-energy processes
Gamma-ray	extreme events	compact objects, particle acceleration

14.1 Resolution

The angular resolution of a telescope is limited by diffraction:

$$\theta \sim 1.22\frac{\lambda}{D},$$

where D is aperture diameter.

Larger telescopes and shorter wavelengths provide better angular resolution.

Diffraction-limited resolution

For a telescope with $D = 2\,\mathrm{m}$ observing visible light at $\lambda = 500\,\mathrm{nm}$,

$$\theta \sim 1.22\frac{5 \times 10^{-7}}{2} \approx 3.1 \times 10^{-7}\,\mathrm{rad}.$$

In arcseconds, multiply by 206265:

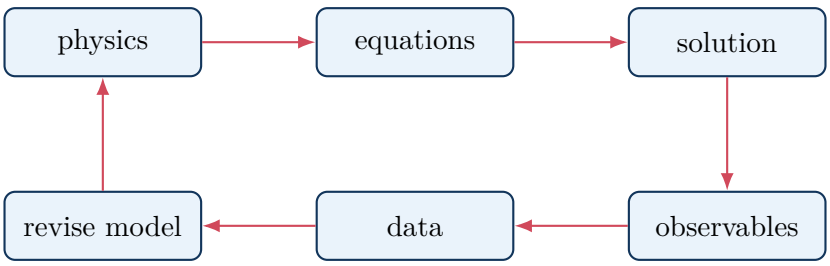
$$\theta \approx 0.064''.$$

15 Astrophysical Modelling and Numerical Thinking

Modern astrophysics is highly computational. Observations are compared with simulations and statistical models.

15.1 A minimal modelling pipeline

1. Define physical assumptions.
2. Write equations.
3. Choose initial and boundary conditions.
4. Solve analytically if possible, numerically if necessary.
5. Convert model predictions into observable quantities.
6. Compare with data.
7. Update assumptions if the model fails.



15.2 Example: numerical orbit update

For a body moving under gravity from mass M, the acceleration is

$$\vec{a} = -\frac{GM}{r^3}\vec{r}.$$

A simple Euler update is

$$\vec{v}_{n+1} = \vec{v}_n + \vec{a}_n \Delta t,$$
$$\vec{r}_{n+1} = \vec{r}_n + \vec{v}_n \Delta t.$$

This method is easy but not very accurate for long-term orbital dynamics. Better methods include symplectic integrators and Runge-Kutta methods.

Numerical warning

A simulation that looks plausible can still be physically wrong. Always check conserved quantities such as energy and angular momentum when appropriate.

16 Exoplanets

An exoplanet is a planet orbiting a star other than the Sun. Exoplanet science connects astrophysics with orbital mechanics, statistics, and data analysis.

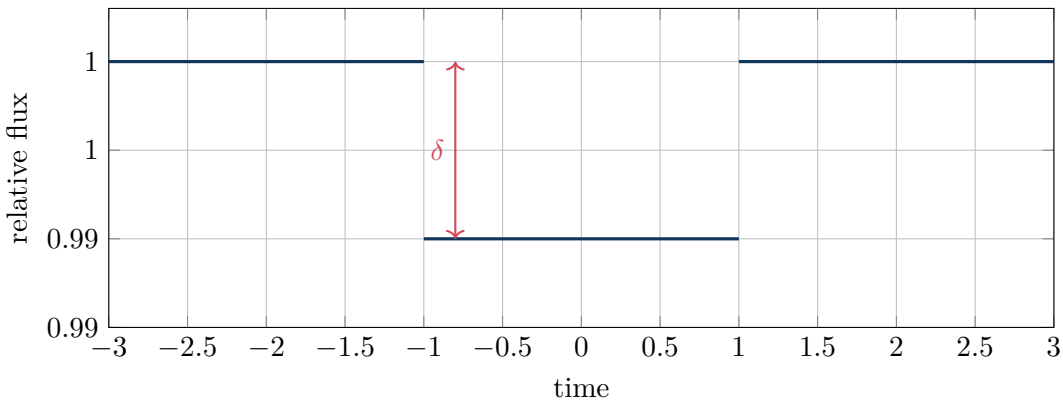
16.1 The transit method

If a planet passes in front of its star, the observed stellar flux decreases slightly. The fractional depth of the transit is approximately

$$\delta \approx \left(\frac{R_p}{R_*}\right)^2,$$

where R_p is planet radius and R_* is stellar radius.

Schematic transit light curve



Planet radius from transit depth

A star similar to the Sun shows a transit depth of $\delta = 0.01$. Then

$$\frac{R_p}{R_*} = \sqrt{0.01} = 0.1.$$

If $R_* = R_\odot$, the planet radius is about $0.1 R_\odot$, comparable to Jupiter.

16.2 The radial velocity method

A planet and star orbit their common center of mass. The star therefore moves slightly, producing Doppler shifts in its spectral lines.

For a circular orbit viewed edge-on, the radial velocity amplitude is roughly

$$K \sim \left(\frac{2\pi G}{P}\right)^{1/3} \frac{M_p}{M_*^{2/3}},$$

when $M_p \ll M_*$. The actual measured quantity includes $\sin i$, where i is orbital inclination.

Inclination degeneracy

Radial velocity alone usually measures $M_p \sin i$, not the true planet mass. Transits help because they imply i is close to $90°$.

16.3 Equilibrium temperature

Ignoring atmosphere and albedo details, a planet's equilibrium temperature can be estimated by energy balance. The absorbed stellar power is approximately

$$\pi R_p^2 (1 - A) \frac{L_*}{4\pi a^2},$$

where A is Bond albedo and a is orbital distance. The emitted thermal power is

$$4\pi R_p^2 \sigma T_{\text{eq}}^4.$$

Equating them gives

$$T_{\text{eq}} = \left(\frac{(1 - A)L_*}{16\pi\sigma a^2}\right)^{1/4}.$$

17 Accretion and High-Energy Astrophysics

Accretion is the process by which matter falls onto a compact object or massive body. It powers some of the most luminous systems in the universe: X-ray binaries, active galactic nuclei, quasars, and some gamma-ray bursts.

17.1 Gravitational energy release

A mass m falling from infinity to radius R around an object of mass M releases energy of order

$$E \sim \frac{GMm}{R}.$$

Per unit mass,

$$\frac{E}{m} \sim \frac{GM}{R}.$$

For compact objects, R is small, so accretion is extremely efficient.

Why accretion can beat nuclear fusion locally

For a neutron star with $M = 1.4M_\odot$ and $R = 12\,\mathrm{km}$,

$$\frac{GM}{R} \approx \frac{6.67 \times 10^{-11} \cdot 2.79 \times 10^{30}}{1.2 \times 10^4} \approx 1.6 \times 10^{16}\,\mathrm{J\,kg^{-1}}.$$

This is a large fraction of c^2, so accretion onto compact objects is a powerful energy source.

17.2 Accretion disks

Matter with angular momentum cannot fall straight inward. It forms a disk. Viscosity and turbulence transport angular momentum outward, allowing mass to spiral inward.

In a thin disk, the orbital speed is approximately Keplerian:

$$\Omega(r) = \sqrt{\frac{GM}{r^3}}.$$

The inner disk becomes hot and can emit X-rays around compact objects.

17.3 Eddington luminosity

Radiation pushes outward on ionized gas by scattering off electrons. Gravity pulls inward. The Eddington luminosity is the luminosity at which outward radiation force balances inward gravity for ionized hydrogen:

$$L_{\mathrm{Edd}} = \frac{4\pi G M m_p c}{\sigma_T},$$

where σ_T is the Thomson scattering cross-section. Numerically,

$$L_{\mathrm{Edd}} \approx 1.3 \times 10^{31} \left(\frac{M}{M_\odot}\right)\,\mathrm{W}.$$

Physical interpretation

The Eddington limit is not an absolute speed limit for luminosity in every geometry. It is a simple spherical estimate. Real systems can exceed it temporarily or anisotropically.

18 Supernovae and Nucleosynthesis

Supernovae are explosive stellar events. They are crucial because they inject energy, momentum, and heavy elements into the interstellar medium.

18.1 Core-collapse supernovae

Massive stars build heavier elements in their cores. Iron is special because fusing iron does not release energy. When the core can no longer support itself, it collapses.

The simplified sequence is:

1. massive star exhausts nuclear fuel in the core;
2. iron core grows and becomes unstable;

3. core collapses to nuclear density;
4. inner core rebounds and neutrino processes help drive an explosion;
5. remnant becomes neutron star or black hole.

18.2 Type Ia supernovae

Type Ia supernovae are associated with thermonuclear disruption of white dwarfs in binary systems. Because their peak luminosities can be standardized, they are important cosmological distance indicators.

18.3 Where elements come from

The broad picture:

- hydrogen and helium mostly formed in the early universe;
- stars produce helium, carbon, oxygen, and heavier elements through fusion;
- supernovae and neutron-star mergers produce many heavy elements;
- stellar winds and explosions return enriched material to space.

Common misconception

The phrase "we are made of stardust" is qualitatively true, but different elements have different origins. Not all heavy elements are produced by the same process.

19 Gravitational Waves and Multi-Messenger Astronomy

Electromagnetic radiation is not the only messenger. Astrophysical information also arrives through neutrinos, cosmic rays, and gravitational waves.

19.1 What is a gravitational wave?

In general relativity, gravity is geometry. Accelerating masses can produce ripples in spacetime curvature called gravitational waves.

A passing gravitational wave changes relative distances by a tiny fractional amount called strain:

$$h \sim \frac{\Delta L}{L}.$$

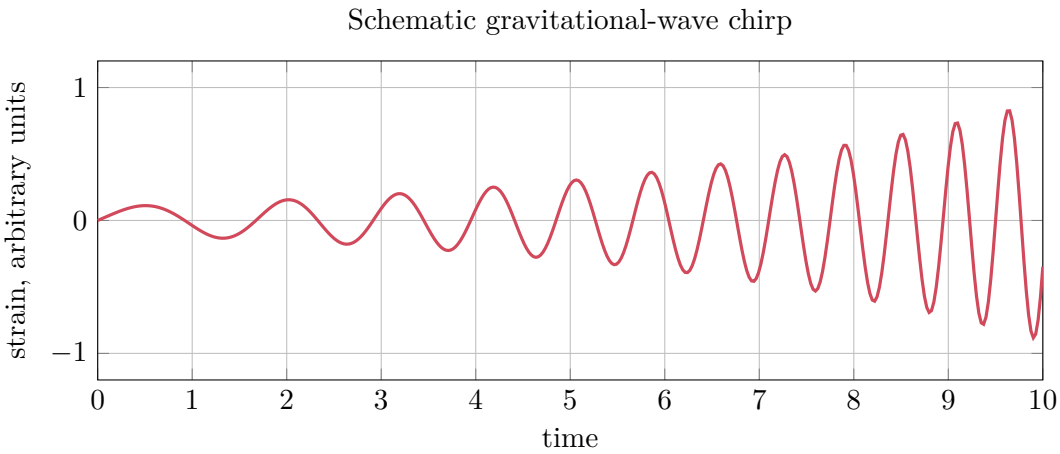
Detected strains are extremely small, often of order 10^{-21}.

19.2 Compact binary inspiral

Two compact objects in orbit emit gravitational waves and lose orbital energy. The orbit shrinks and the frequency increases, producing a chirp signal.

The characteristic frequency is related to orbital frequency. Roughly,

$$f_{\mathrm{GW}} \approx 2 f_{\mathrm{orb}}.$$



19.3 Multi-messenger reasoning

A single event can be observed in multiple channels:

- gravitational waves reveal masses, spins, and orbital dynamics;
- electromagnetic light reveals ejecta, environment, and nucleosynthesis;
- neutrinos reveal processes hidden from photons.

The scientific value comes from consistency: different messengers constrain the same physical event.

20 Data Analysis in Astrophysics

Astrophysical conclusions are statistical. Observations have noise, selection effects, calibration uncertainties, and model degeneracies.

20.1 Signal-to-noise ratio

A basic measure of detection significance is signal-to-noise ratio:

$$\mathrm{SNR} = \frac{\text{signal}}{\text{noise}}.$$

For photon counting, noise is often approximately Poissonian. If N photons are counted, then the shot noise is roughly

$$\sigma_N \approx \sqrt{N}.$$

Thus

$$\mathrm{SNR} \approx \frac{N}{\sqrt{N}} = \sqrt{N}.$$

Collecting more photons improves SNR only as the square root of exposure.

Exposure time and SNR

If an observation detects $N = 10^4$ photons, the photon-limited SNR is about

$$\sqrt{10^4} = 100.$$

To double the SNR to 200, one needs four times as many photons, not twice as many.

20.2 Model fitting

Suppose data points (x_i, y_i) have uncertainties σ_i, and a model predicts $y_{\text{model}}(x_i; \theta)$. A common fitting statistic is

$$\chi^2(\theta) = \sum_i \left(\frac{y_i - y_{\text{model}}(x_i; \theta)}{\sigma_i} \right)^2 .$$

The best-fitting parameters often minimize χ^2, but interpretation requires care.

20.3 Selection effects

A survey does not observe all objects equally. Brighter objects are easier to detect; nearby objects are overrepresented; some wavelengths are blocked by dust or atmosphere.

Selection bias

A catalog is not the universe. It is the universe filtered through instrumental limits, observing strategy, and detection algorithms.

20.4 Degeneracy

Two different physical models can sometimes produce similar observations. For example, a red color may indicate a cool star, dust reddening, or redshift. Astrophysics often requires combining independent measurements to break degeneracies.

21 Worked Problems

21.1 Problem 1: Temperature from wavelength peak

Wien's displacement law is

$$\lambda_{\text{max}} T = b,$$

where $b \approx 2.90 \times 10^{-3}\,\text{m}\,\text{K}$.

A star has peak wavelength $\lambda_{\text{max}} = 500\,\text{nm}$. Estimate its temperature.

Solution. Convert wavelength:

$$500\,\text{nm} = 5.0 \times 10^{-7}\,\text{m}.$$

Then

$$T = \frac{b}{\lambda_{\text{max}}} = \frac{2.90 \times 10^{-3}}{5.0 \times 10^{-7}} = 5.8 \times 10^3\,\text{K}.$$

This is approximately solar surface temperature.

21.2 Problem 2: Radius from luminosity and temperature

A star has $L = 16 L_\odot$ and $T = 2T_\odot$. Find $R/R_\odot$.

Solution. Use

$$\frac{L}{L_\odot} = \left(\frac{R}{R_\odot}\right)^2 \left(\frac{T}{T_\odot}\right)^4 .$$

Substitute:

$$16 = \left(\frac{R}{R_\odot}\right)^2 2^4 = 16\left(\frac{R}{R_\odot}\right)^2.$$

Thus

$$\frac{R}{R_\odot} = 1.$$

The star is hotter than the Sun but has the same radius, so it is more luminous.

21.3 Problem 3: Schwarzschild radius

Find the Schwarzschild radius of a $10M_\odot$ black hole.

Solution. The Schwarzschild radius scales linearly with mass:

$$r_s = \frac{2GM}{c^2}.$$

For one solar mass, $r_s \approx 3\,\mathrm{km}$. Therefore for $10M_\odot$,

$$r_s \approx 30\,\mathrm{km}.$$

21.4 Problem 4: Galaxy mass from rotation

A galaxy has flat rotation speed $v = 220\,\mathrm{km\,s^{-1}}$ out to radius $r = 20\,\mathrm{kpc}$. Estimate mass inside that radius.

Solution. Use

$$M = \frac{v^2 r}{G}.$$

Convert units:

$$v = 2.2 \times 10^5\,\mathrm{m\,s^{-1}},$$

$$r = 20 \times 10^3\mathrm{pc} = 2.0 \times 10^4(3.09 \times 10^{16})\,\mathrm{m} = 6.18 \times 10^{20}\,\mathrm{m}.$$

Then

$$M = \frac{(2.2 \times 10^5)^2(6.18 \times 10^{20})}{6.67 \times 10^{-11}} \approx 4.5 \times 10^{41}\,\mathrm{kg}.$$

In solar masses:

$$M \approx \frac{4.5 \times 10^{41}}{1.99 \times 10^{30}} \approx 2.3 \times 10^{11} M_\odot.$$

22 Common Misconceptions

Misconception 1: Bright means close

A bright object may be close and faint, or far and luminous. Flux alone does not give luminosity without distance.

Misconception 2: A red star is always faint

Red means relatively cool surface temperature, not necessarily low luminosity. A red giant can be extremely luminous because its radius is huge.

Misconception 3: Black holes suck everything in

Far from a black hole, gravity behaves like gravity from any object with the same mass. A black hole is dangerous because it can be extremely compact, not because it has magical extra gravity at large distances.

Misconception 4: Dark matter is just ordinary matter we cannot see

Some unseen ordinary matter exists, but multiple lines of evidence indicate that most dark matter is not ordinary baryonic matter.

23 Exercises

A. Basic calculations

1. A star has flux $F = 1.0 \times 10^{-12}\,\mathrm{W\,m^{-2}}$ and luminosity $L = L_\odot$. Estimate its distance.
2. A spectral line at 600 nm is observed at 601.2 nm. Estimate radial velocity.
3. A star has $R = 2R_\odot$ and $T = 3T_\odot$. Find $L/L_\odot$.
4. Estimate the Schwarzschild radius of a $4 \times 10^6 M_\odot$ black hole.
5. A galaxy has $v = 180\,\mathrm{km\,s^{-1}}$ at $r = 15\,\mathrm{kpc}$. Estimate the mass inside this radius.

B. Conceptual questions

1. Why does a hotter blackbody peak at shorter wavelength?
2. Why does hydrogen fusion require high temperature?
3. Why can a white dwarf be hot but faint?
4. Why do flat galaxy rotation curves suggest dark matter?
5. Why is the cosmic microwave background important evidence for a hot early universe?

24 Formula Sheet

Light and distance

$$F = \frac{L}{4\pi d^2}$$

$$m - M = 5\log_{10}\left(\frac{d}{10\mathrm{pc}}\right)$$

$$\frac{\Delta\lambda}{\lambda_0} \approx \frac{v_r}{c}$$

Blackbody

$$L = 4\pi R^2 \sigma T^4$$

$$\lambda_{\max} T = b$$

Gravity

$$F = G\frac{m_1 m_2}{r^2}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$P^2 = \frac{4\pi^2}{GM} r^3$$

$$v_{\mathrm{esc}} = \sqrt{\frac{2GM}{R}}$$

$$r_s = \frac{2GM}{c^2}$$

Stars

$$\frac{\mathrm{d}m}{\mathrm{d}r} = 4\pi r^2 \rho$$

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{Gm\rho}{r^2}$$

$$\frac{\mathrm{d}L}{\mathrm{d}r} = 4\pi r^2 \rho \epsilon$$

$$1 + z = \frac{a(t_0)}{a(t_{\mathrm{emit}})}$$

Cosmology

$$v = H_0 d$$

$$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}$$

25 Suggested Learning Path

For a first course, study the material in this order:

1. Understand flux, luminosity, and distance.
2. Learn blackbody radiation and spectra.
3. Review Newtonian gravity and circular orbits.
4. Study the H-R diagram and stellar classification.
5. Learn the basic stellar structure equations.
6. Study stellar evolution and compact remnants.
7. Move to galaxies, rotation curves, and dark matter.
8. Finish with expansion, redshift, and the basic cosmological model.

Final perspective

Astrophysics is not a list of disconnected facts about space. It is the systematic use of physics to infer hidden properties of distant systems from limited information, mostly light.